UVEK background in Premise

Review snapshot | 10 November 2026

Development paused at the user's request. Expanded goal incomplete.

What is ready to use

The normal public workflow supports **biomass, electricity and heat** for the reviewed REMIND SSP2-NPi 2050 scenario, using UVEK 2026 cut-off with the ecoinvent 3.10 biosphere. The combined export contains **12,622 activities**, with automatic CLIC selection, complete linked readback and retained audits/provenance. This is screening support for the reviewed scenario, not qualification of every model or technology.

Area	Evidence at the pause
Electricity pilot	All 42 original deferrals have a final disposition: 39 fully restored, 2 partial mappings and 1 final unsupported exchange; none pending.
Public three-sector workflow	Biomass + electricity + heat: normal API, certification and Brightway export verified. Expanded selection: 12 roots / 76 source records.
Emissions	Internal callback and mapping batches verified through commit 2528051f. Public emissions update/export remains gated.
Broader sector scope	3 of 19 updates operational for the reviewed scenario. Emissions and 15 other updates remain unqualified end to end; 11 heat-role gaps remain.

What the mapping milestone proves

The goal-starting 20 electricity deferrals ended as **17 restored, 2 partially restored and 1 final unsupported**. The original nine roots and 55 source pins remain a regression baseline. Signed physical quantities, supported uncertainty, foreground ownership and complete links were checked. Cut-off/consequential conversion remains excluded.

Latest verified internal emissions component

3,487 inputs in 2,816 callers; 289 owned combustion variants; 483 gaseous-air changes. The latest qualification passed 61 focused tests and **196 signed finite LCI checks**. The isolated wheel matched all 282 resources and 131 runtime modules, and reproduced graph, metadata and 3,105 provenance events. Earlier variants retained their scientific coordinates on identical input data.

These checks establish bounded implementation and numerical behavior. They do not prove LCIA equivalence, impact completeness or a qualified public four-sector export.

Implementation boundary

Recent machinery and gas-supply batches changed YAML aliases and existing JSON review data, with no Python edits. Earlier work added family dispatch, validation/provenance and source-cache/export-ownership fixes. All commits are local; nothing was pushed. Licensed inventories, keys and private analysis records are not bundled in this review.

Scientific limits and review priorities

Keep the verified baseline separate from the unfinished work.

Choices that remain visible in results

Signs and reference bases. A negative treatment input is not, by itself, an avoided burden. Its sign must be interpreted with the supplier's production convention, allocation and functional reference. No absolute-value repair or invented conversion factor is justified.

Proxy and omission boundaries. Geographic, technology, allocation and material-basis approximations remain screening assumptions with unquantified score effects. Two paperboard mappings retain final omitted branches; the unsupported GOES exchange and separate CSP used-nitrate reuse-credit omission remain audited. Upstream biosphere exclusions also prevent an impact-completeness claim.

Emissions controls. Gas compression/process energy uses an explicit industrial-gas screening proxy. Two unresolved LBFAC32-emitter inputs use World factors without relocating source inventories. Caller leakage, CO2, particles, non-air releases, aggregate VOC and unreviewed species remain unchanged; industrial and energy-industry control regimes are not established equivalent.

Unfinished batch: not qualified for use

The paused draft adds **137 input entries across 25 source inventories** in four mapping files. Its 61 focused tests pass, but the broader signed-inventory check **failed the numerical-residual criterion**. The cause has not been diagnosed. These edits are uncommitted and are excluded from the notebook's frozen baseline. Two additional lean-burn inputs remain unchanged because source methane combines combustion and distribution leakage.

Decisions for the review

1. Release scope. Decide whether the next deliverable should consolidate the working three-sector route or pursue broader sector support first. The original expanded goal remains intact but paused.

2. Scientific acceptance. Agree which screening proxies/omissions are acceptable for the intended use and what score-sensitivity evidence is needed before quantitative conclusions.

3. Re-entry criteria. Before opening emissions: resolve mixed-mechanism/direct/foreground and particle/VOC coverage, investigate the draft residual failure, and pass normal public combined certification/export and representative readback calculations.

Run the review notebook

examples/uvek_2026_review_workflow.ipynb defaults to read-only inspection of the verified existing export. It checks the frozen package, complete links and signed inventories. Explicit switches enable a fresh public build, a new database export or a separate internal emissions preview. An encrypted IAM file and interactive key are required only for a fresh build. No credentials or licensed inventories are embedded.

Audit trail: uvek-mapping-decisions.md; uvek-sector-progress.json; electricity-milestone, heat-public and emissions-gas-supply verification JSON files under docs/development. Local mapping commits: 2528051f (gas supply), a6e32c6c (machinery), 0d200278 (heating), f877976b (callback). Earlier heat evidence reports 13 gaps; two later fuel-cell bindings reduce the current count to 11.